

PROJECT REPORT



SIMULATIONS OF LARGE- N MATRIX MODELS USING HAMILTONIAN MONTE CARLO

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ABSTRACT

In this project, we simulate a non-abelian matrix model using the method of Hamiltonian Monte Carlo (HMC). Here, first we discuss Monte Carlo method of sampling and Markov Chain Monte Carlo (MCMC) method using Hamiltonian dynamics. Then we apply HMC method to simulate the matrix model.

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I owe my sincere gratitude to Dr. Anosh Joseph, for providing me an opportunity to do the project work and giving me all support and guidance, which made me complete the project duly. I am very grateful to my friend Nikhil Tanwar, for helping me in every difficulty regarding C++ programming I faced.

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1 Monte Carlo Method

Monte Carlo method¹ is a tool that uses random numbers to compute quantities that are not random.

Let X be a random variable with expectation value

$$A = E[X]. \quad (1)$$

If $X_1, X_2, \dots, X_n$, are n independently created random variables with the same distribution, then the approximate value of expectation is written as

$$A \approx \bar{A}_n = \frac{1}{n} \sum_{k=1}^n X_k. \quad (2)$$

Suppose $X \in R^n$ is an n -component random vector and $B \subseteq R^n$ is an event and $p(x)$ be the probability density function, then we can write

$$Pr(B) = \int_{x \in B} p(x) dx. \quad (3)$$

The expected value is

$$\mu_k = E[X_k] = \int_{R^n} x_k p(x) dx. \quad (4)$$

1.1 Integrals Using Monte Carlo

1.1.1 Stratified Sampling

In order to compute integral with this method we divide the integration domain R into k subregions R_i for $i = 1, \dots, k$.

$$E[x] = \int_R x p(x) dx = \sum_{j=1}^k \int_{R_j} x p(x) dx. \quad (5)$$

So, the MC estimate of the expectation value is

¹See Ref. [4] for an introductory tutorial on Monte Carlo methods.

$$I = \sum_{j=1}^k \frac{Vol(R_j)}{n_j} \sum_{i=1}^{n_j} x_i. \quad (6)$$

Here n_j is the number of points used for Monte Carlo (MC) integration on R_j , and $Vol(R_j)$ is the sub-domain volume.

1.1.2 Importance Sampling

In the integral, $p(x)$ is the probability distribution function (PDF), and it may not necessarily be the best PDF for MC integration. In that case, we try to use a different and simpler PDF say, $q(x)$, from which we can quickly draw the samples. So, we call $q(x)$ as *importance density* or *proposal distribution*. Then we can write

$$\begin{aligned} E[y] &= E[f(x)] = \int_b^a f(x)p(x)dx, \\ &= \int_b^a \frac{f(x)p(x)}{q(x)}q(x)dx, \\ &= E\left[\frac{f(x)p(x)}{q(x)}\right]. \end{aligned} \quad (7)$$

Now by obtaining n samples $x_i \sim q(x)$, for $i = 1, \dots, n$, the estimate comes out to be

$$\begin{aligned} I &= \frac{1}{n} \sum_{i=1}^n \frac{f(x_i)p(x_i)}{q(x_i)}, \\ &= \frac{1}{n} \sum_{i=1}^n W(x_i)f(x_i). \end{aligned} \quad (8)$$

Here $W(x_i) = \frac{p(x_i)}{q(x_i)}$ are the importance weights.

2 Markov Chain Monte Carlo (MCMC)

MCMC is a method that is used to generate samples from a general probability distribution function $p(x)$ from which it is tough to sample directly. The approach that we use here is to use a sequence of samples x_k from a Markov chain, so as the sequence converges on $p(x)$ as $k \rightarrow \infty$.

2.1 Markov Chains

A Markov process is a procedure that uses a probabilistic approach to find the future state from the current state provided these generated states follow the Markov property. A process is said to have Markov property if all the past states depend only on the current state. So, we write the conditional probability distribution of the future states as

$$p(x_{k+1}|x_0, x_1, \dots, x_k) = p(x_{k+1}|x_k). \quad (9)$$

A series of random variables x_k generated by a Markov process is called Markov chain. The random variable x_k denotes the state of that process at time k . Probability of transition from a state x to a state y is called *transition probability density* and is denoted by $p(y|x)$. It satisfies

$$\int p(y|x)dy = 1. \quad (10)$$

A unique steady-state distribution π independent of the initial state exists if a Markov chain is ergodic

$$\pi(y) = \int p(y|x)\pi(x)dx. \quad (11)$$

2.2 Detailed Balance Property

This property assures the invariance of π under a transition

$$p(x|y)\pi(y) = p(y|x)\pi(x), \forall x, y. \quad (12)$$

So, while performing MCMC sampling, the samples are produced by a homogeneous, reversible, ergodic Markov chain with invariant distribution π .

2.3 Metropolis Algorithm

Suppose we want to draw samples from the target density $\pi(x)$, for which the normalizing factor is known up to a constant Z or might be unknown

$$\pi(x) = \frac{\pi(x)}{Z}. \quad (13)$$

We choose a proposal density $q(y|x)$ that is symmetric with the property

$$\int q(y|x)dx = 1. \quad (14)$$

To generate a Markov chain, which converges to $\pi(x)$ we need to take the following steps.

1. Initiate the state. That is $X_1 = x_1$
2. Iterate for $t = 1, 2, \dots$
 - Sample y from $q(y|x_t)$. Think of y as a *proposed value* for x_{t+1} .
 - Compute

$$A = \min \left(1, \frac{\pi(y)}{\pi(x_t)} \right). \quad (15)$$

A is called the *acceptance probability*.

- With probability A “accept” the proposed value, and set $x_{t+1} = y$. Otherwise set $x_{t+1} = x_t$.

3 MCMC using Hamiltonian Dynamics

Hamiltonian Monte Carlo (HMC) is a technique that uses Hamiltonian dynamics to provide distant proposals for the Metropolis scheme. As a result one can avoid the slow exploration of the state space that results from simple random-walk method.

3.1 Hamiltonian Dynamics

A d -dimensional momentum vector, p , and d -dimensional position vector, q , together with forms a state-space (q, p) of dimensions $2d$. Hamiltonian dynamics functions on this $2d$ dimensional state space. Hamiltonian, $H(q, p)$, is an operator described by the functions of q and p .

3.1.1 Equations of Motion

The rate of change of p and q over time, t , is determined by the partial derivatives of the Hamiltonian. The equations describing these relations are called Hamilton’s equations and are given as

$$\frac{dq_i}{dt} = \frac{\partial H}{\partial p_i}, \quad (16)$$

$$\frac{dp_i}{dt} = -\frac{\partial H}{\partial q_i}, \quad (17)$$

for $i = 1, \dots, d$.

These equations represents a mapping, T_s , from a state at any time t to a state at time $t + s$. T_s is assumed to be independent of t .

3.1.2 Potential and Kinetic Energies

For the Hamiltonian Monte Carlo, we ordinarily use Hamiltonian functions that can be written as follows

$$H(q, p) = U(q) + K(p). \quad (18)$$

Here, $U(q)$ and $K(p)$ are called potential and kinetic energy, respectively. $U(q)$ is defined as minus the log probability density of the distribution for q , and $K(p)$ is defined as

$$K(p) = \frac{p^T M^{-1} p}{2}, \quad (19)$$

where M is called a mass matrix. It is a positive-definite, symmetric diagonal matrix, and is usually a scalar multiple of the identity matrix.

Assuming H and K of these forms, we can write new forms of Hamilton's equation

$$\frac{dq_i}{dt} = [M^{-1}p]_i, \quad (20)$$

$$\frac{dp_i}{dt} = -\frac{dU}{dq_i} \quad (21)$$

$$(22)$$

where i runs from 1 to d .

3.1.3 A One-dimensional Example

Suppose we have a Hamiltonian, defined as

$$H(q, p) = U(q) + K(p), \quad (23)$$

$$U(q) = \frac{q^2}{2}, \quad (24)$$

$$K(p) = \frac{p^2}{2}. \quad (25)$$

Then, the equations of motion are given as

$$\frac{dq}{dt} = p, \quad (26)$$

$$\frac{dp}{dt} = -q. \quad (27)$$

Solutions have the form

$$q(t) = r \cos(a + t), \quad (28)$$

$$p(t) = -r \sin(a + t). \quad (29)$$

Henceforth the mapping T_s is a rotation by s radians clockwise around the origin in the (q, p) plane.

3.2 Properties of Hamiltonian Dynamics

1. **Reversibility.** Hamiltonian dynamics is reversible. The mapping T_s that maps a state $(q(t), p(t))$ at time t to a state $(q(t + s), p(t + s))$ at time $t + s$ is one-to-one and therefore has its inverse T_{-s} . For the above 1d example T_{-s} is counterclockwise rotation.

2. **Conservation of the Hamiltonian.** Hamiltonian is conserved. This can be shown as

$$\begin{aligned} \frac{dH}{dt} &= \sum_{i=1}^d \left[\frac{dq_i}{dt} \frac{dH}{dq_i} + \frac{dp_i}{dt} \frac{dH}{dp_i} \right] \\ &= \sum_{i=1}^d \left[\frac{\partial H}{\partial p_i} \frac{\partial H}{\partial q_i} - \frac{\partial H}{\partial q_i} \frac{\partial H}{\partial p_i} \right] \\ &= 0. \end{aligned} \quad (30)$$

3. **Volume preservation.** It preserves the volume in (q, p) space. Let T_s defines a map to the points in some region R of (q, p) space, with volume V , then the image of R under T_s will also have volume V .

4. **Symplecticness.** Let $z = (q, p)$, the symplecticness condition is that the Jacobian matrix, B_s , of the mapping T_s satisfies

$$B_s^T J^{-1} B_s = J^{-1}, \quad (31)$$

where

$$J = \begin{vmatrix} 0_{d \times d} & I_{d \times d} \\ -I_{d \times d} & 0_{d \times d} \end{vmatrix}. \quad (32)$$

This definition also implies volume conservation.

3.3 Discretization of Hamilton's Equations

For computational purposes we discretize Hamilton's equations. We use Leapfrog method to discretize them.

We assume Hamiltonian of the form $H(q, p) = U(q) + K(p)$. For simplicity, we assume kinetic energy of the form

$$K(p) = p^T M^{-1} p, \quad (33)$$

and further M is diagonal with diagonal elements $m_1, m_2, \dots, m_d$. So, we can write

$$K(p) = \sum_{i=1}^d \frac{p_i^2}{2m_i}. \quad (34)$$

3.3.1 Euler's Method

One way to discretize Hamilton's equation is using Euler's method.

In the discretized form we have

$$\begin{aligned} p_i(t + \epsilon) &= p_i(t) + \epsilon \frac{dp_i}{dt}(t) = p_i(t) - \epsilon \frac{\partial U}{\partial q_i}(q(t)), \\ q_i(t + \epsilon) &= q_i(t) + \epsilon \frac{dq_i}{dt}(t) = q_i(t) + \epsilon \frac{p_i(t)}{m_i}, \end{aligned} \quad (35)$$

where i runs from 1 to d .

Suppose we start with given values of $q_i(0)$ and $p_i(0)$ at $t = 0$, to get a trajectory of position and momentum values, we iterate the above equations for step sizes $\epsilon, 2\epsilon, 3\epsilon, \dots$. Thus, we find (approximate) values for $q(\tau)$ and $p(\tau)$ after τ/ϵ steps (assuming that τ/ϵ is an integer).

In Fig. 1 we show how the trajectory for momentum and position computed for Hamiltonian,

$$H = \frac{p^2}{2} + \frac{q^2}{2}, \quad (36)$$

using the Euler method diverges to infinity given that the actual trajectory is circular.

3.3.2 Modified Euler's Method

To eliminate divergence, we slightly modify Euler's method, as follows

$$p_i(t + \epsilon) = p_i(t) - \epsilon \frac{\partial U}{\partial q_i}(q(t)), \quad (37)$$

$$q_i(t + \epsilon) = q_i(t) + \epsilon \frac{p_i(t + \epsilon)}{m_i}. \quad (38)$$

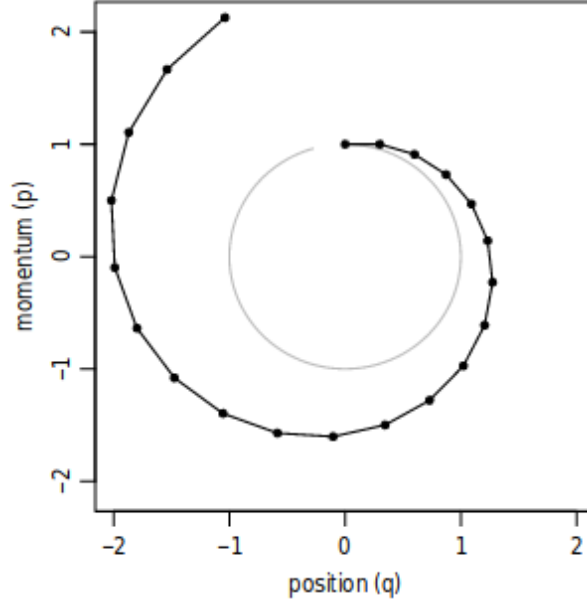


Figure 1: The Euler's method for Hamiltonian $H = \frac{p^2}{2} + \frac{q^2}{2}$ with stepsize $\epsilon = 0.3$, and initial conditions $q(0) = 0$ and $p(0) = 1$. This figure is taken from Ref. [3].

While using this method we compute the value of momentum and then use this new value of momentum to compute the new value of position. We compute in the same way for each component of position and momentum, indexed by $i = 1, \dots, d$.

Fig. 2 shows the trajectory using the Modified Euler's method.

3.3.3 Leapfrog Method

To get more improved results, we further modify Euler's method, as follows

$$p_i(t + \frac{\epsilon}{2}) = p_i(t) - \frac{\epsilon}{2} \frac{\partial U}{\partial q_i}(q(t)), \quad (39)$$

$$q_i(t + \epsilon) = q_i(t) + \epsilon \frac{p_i(t + \frac{\epsilon}{2})}{m_i}, \quad (40)$$

$$p_i(t + \epsilon) = p_i(t + \frac{\epsilon}{2}) - \frac{\epsilon}{2} \frac{\partial U}{\partial q_i}(q(t + \epsilon)). \quad (41)$$

We start with a half step for a random value of momentum variable, then using the new value of the momentum variable, we make a full step for the position variable and lastly, using the new value for the position variable, we make another half step for the momentum variable.

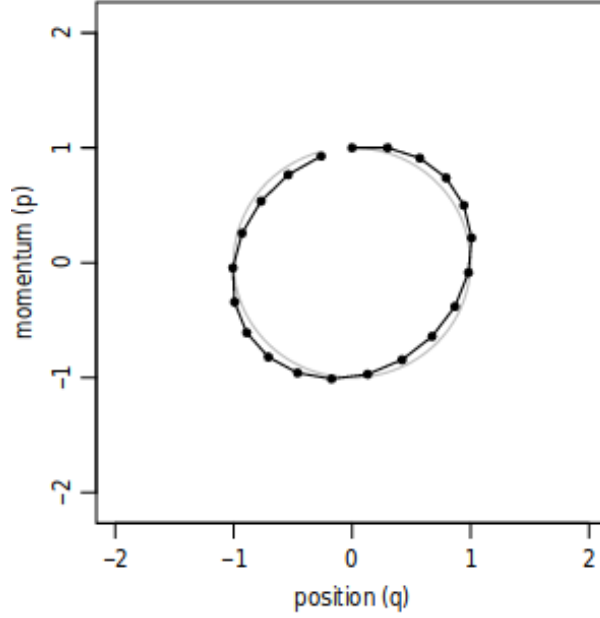


Figure 2: The modified Euler's method for Hamiltonian $H = \frac{p^2}{2} + \frac{q^2}{2}$, with stepsize $\epsilon = 0.3$, and initial conditions $q(0) = 0$ and $p(0) = 1$. This figure is taken from Ref. [3].

Fig. 3 shows the trajectory using the Leapfrog method.

3.3.4 Errors in Discretization Methods

The discretization methods we discussed above have local and global errors. The error that is generated after making one step i.e., moving from t to $t + \epsilon$ is called the local error, and the error that is generated after iterating for some fixed time interval is called the global error.

The Euler method and its modification, both have order ϵ^2 local error and order ϵ global error. The leapfrog method has order ϵ^3 local error and order ϵ^2 global error.

3.4 Canonical Distributions

The potential energy can be related to the distribution we wish to sample using the canonical distribution. The canonical distribution over some states has probability density

$$P(x) = \frac{1}{Z} \exp(-E(x)/T). \quad (42)$$

where $E(x)$ represents energy of a state x .

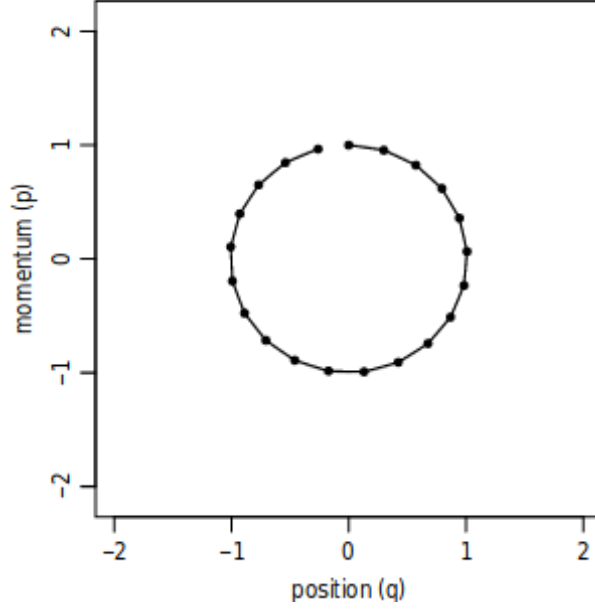


Figure 3: Leapfrog method for Hamiltonian $H = \frac{p^2}{2} + \frac{q^2}{2}$ with stepsize $\epsilon = 0.3$, and initial conditions $q(0) = 0$ and $p(0) = 1$. This figure is taken from Ref. [3].

Hamiltonian is an energy function of state (q, p) . So, if we have $H(q, p) = U(q) + K(p)$ then

$$P(q, p) = \frac{1}{Z} \exp(-U(q)/T) \exp(-K(p)/T). \quad (43)$$

We take $T = 1$ and define potential energy as

$$U(q) = -\ln(\pi(q)L(q|D)), \quad (44)$$

where $\pi(q)$ represents the prior density, and $L(q|D)$ represents the likelihood function given data D .

3.5 The HMC Algorithm

We set $T = 1$ and let Hamiltonian and kinetic energy of the form

$$H(q, p) = U(q) + K(p), \quad (45)$$

$$K(p) = \sum_{i=1}^d \frac{p_i^2}{2m_i}. \quad (46)$$

This drives p to have a zero-mean Gaussian distribution with each component p_i having variance equal to m_i .

3.5.1 The Two Steps of the HMC Algorithm

1. In this step, new values of momentum variables are extracted randomly from their Gaussian distribution, independent of any current values of the position variables. Gaussian distribution for the above defined kinetic energy has mean zero and variance m_i for each component p_i .
2. In this step, Metropolis update is made. First, Hamilton's equations are iterated for L steps using the Leapfrog method, with certain stepsize ϵ . To obtain better performance, we set parameters L and ϵ accordingly. At the end of the trajectory, we negate the momentum variables to obtain proposed state (q^*, p^*) . This obtained state is accepted as the next state of the Markov chain with acceptance probability

$$\min[1, e^{(-H(q^*, p^*) + H(q, p))}] = \min[1, e^{(-K(p^*) + K(p) - U(q^*) + U(q))}]. \quad (47)$$

If the proposed state is rejected, then the next state is set the same as the current state. Momentum variables are negated at the end of the trajectory to make the Metropolis scheme symmetrical, which is needed for validation of the acceptance probability.

3.6 Results from the One-dimensional Example

In Figs. 4 and 5 we give the plots of trajectories for the Hamiltonian of the type given in Sec. 3.1.3.

4 Large- N Reduced Model of Yang-Mills Theory

The type IIB matrix model, is the reduced model of 10-dimensional supersymmetric Yang-Mills theory and is anticipated to be a nonperturbative formulation of super-string theory.

Here, we study large- N reduced model of Yang-Mills theory, which we also call as the "bosonic" matrix model. It is the bosonic part of the type IIB matrix model.

The action of this model is given as

$$S = -\frac{1}{4g^2} \text{Tr}([A_\mu, A_\nu]^2), \quad (48)$$

where g is a coupling parameter and A_μ ($\mu = 1, \dots, D$) are $N \times N$ hermitian traceless matrices and correspond to the space-time coordinates.

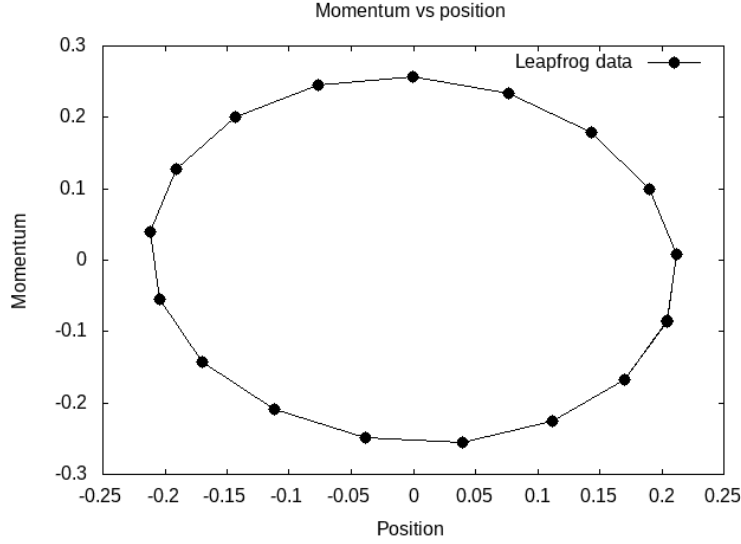


Figure 4: Position against momentum using HMC algorithm for the Hamiltonian $H = \frac{p^2}{2} + \frac{q^2}{2}$. We used step size $\epsilon = 0.3$.

Here, we study the dynamical details of the model related to the eigenvalues of these A_μ matrices. One important quantity in this model, which is the extent of the space-time, is defined by

$$R = \sqrt{\left\langle \frac{1}{N} \text{Tr}(A^2) \right\rangle}. \quad (49)$$

We also study the quantities which represent the uncertainty of the space-time coordinates and an order parameter for the spontaneous symmetry breaking (SSB) of the Lorentz invariance.

4.1 The Extent of Spacetime

For $D > 2$ and large N , the integral of A_μ is convergent due to an attractive potential between the eigenvalues of A_μ . So, we rescale A_μ as $A'_\mu = A_\mu / \sqrt{g}$, where g acts as a scale parameter. Now, since quantity R is proportional to $\sqrt{g}$ so, parametrized form of R is $\sqrt{g}N^\omega$ with ω playing an important role to determine the space-time dimension in IIB model and is determined from the eigenvalue distribution of A_μ .

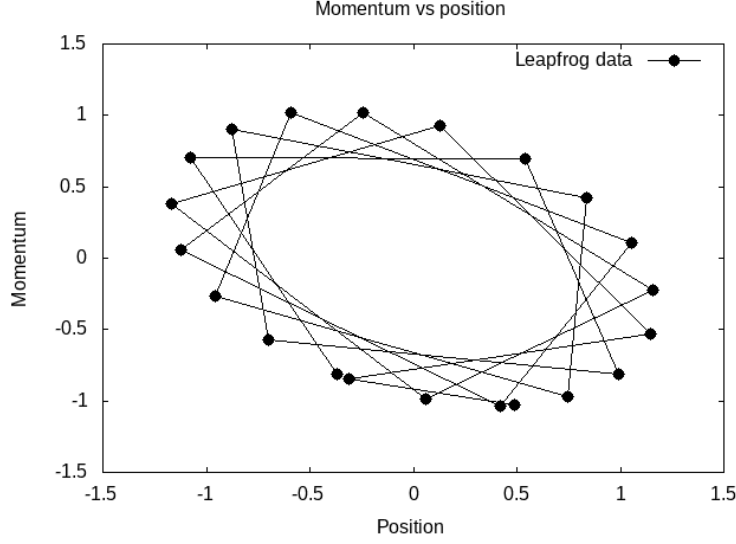


Figure 5: Position against momentum using HMC algorithm for the Hamiltonian $H = \frac{p^2}{2} + \frac{q^2}{2}$. We used step size $\epsilon = 1.2$.

We can define the following observables in the model

$$(a) : \left\langle \frac{1}{N} \text{Tr}(A^2) \right\rangle \times (\sqrt{g} N^{1/4})^{-2}, \quad (50)$$

$$(b) : \left\langle \frac{1}{N} \text{Tr}((A_\mu A_\nu)^2) \right\rangle \times (\sqrt{g} N^{1/4})^{-4}, \quad (51)$$

$$(c) : \left\langle \frac{1}{N} \text{Tr}((A^2)^2) \right\rangle \times (\sqrt{g} N^{1/4})^{-4}, \quad (52)$$

$$(d) : \left\langle \frac{1}{N} \text{Tr}(F^2) \right\rangle \times (\sqrt{g} N^{1/4})^{-4}, \quad (53)$$

where $F_{\mu\nu} = i[A_\mu, A_\nu]$.

Above quantities are normalized so, they are dimensionless and hence, independent of g . Quantity (d) can also be obtained analytically as

$$(d) = D(1 - 1/N^2). \quad (54)$$

In Figs. 6 and 7, we show plots of the above defined quantities (a) – (d) for $D = 4, 10$ with $N = 4, 8, 16, 32, 64, 128, 256$. Important thing to note is the quantities (a) – (d) are constant for $N \geq 16$ and the quantity R is found to be of order $O(\sqrt{g} N^{1/4})$.

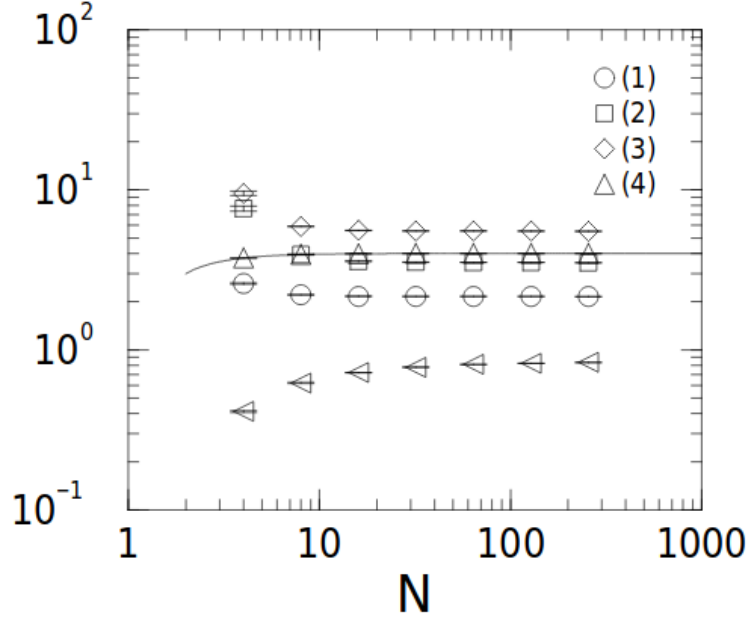


Figure 6: The quantities (a) – (d) are plotted for $D = 4$. The solid line represents the exact result for quantity (d). The tilted triangles are the plot of $\langle \Delta^2 \rangle / (\sqrt{g} N^{1/4})^2$, which is discussed in Sec. 4.2. This figure is taken from Ref. [1].

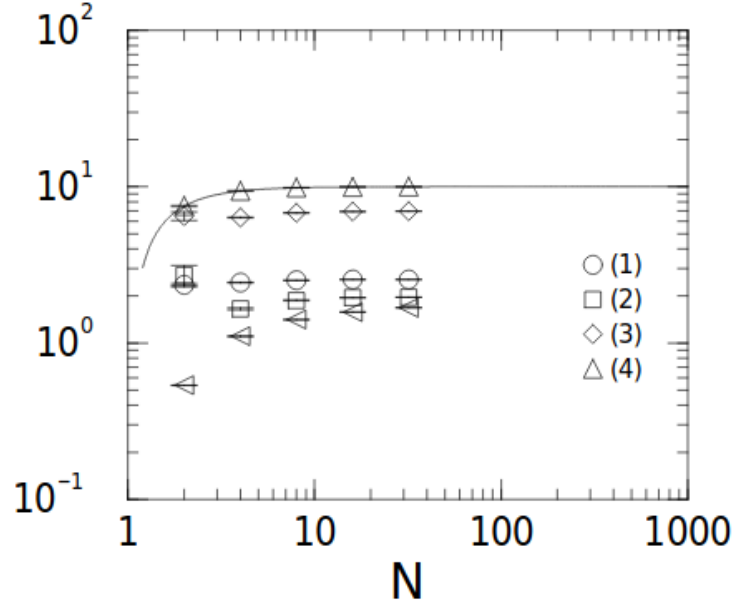


Figure 7: The quantities (a) – (d) are plotted for $D = 10$. The solid line represents the exact result for quantity (d). The tilted triangles are the plot of $\langle \Delta^2 \rangle / (\sqrt{g} N^{1/4})^2$, which is discussed in Sec. 4.2. This figure is taken from ref. [1].

4.2 Breakdown of the Classical Spacetime Picture

In the type IIB matrix model, the eigenvalues of A_μ matrices represent the space-time coordinates. If these matrices are commuting, then they can be diagonalized, and the diagonal elements represent the classical space-time coordinate. Therefore, to estimate the extent to which the classical space-time picture is broken, a quantity is defined that measures the uncertainty of space-time coordinates. The expression for this quantity is given as

$$\Delta = \min_{\{e_i\}} \left(\frac{1}{N} \sum_i \delta(i)^2 \right). \quad (55)$$

Note that $\Delta = 0$ if and only if A_μ are diagonalizable simultaneously.

In Figs. 6 and 7, the tilted triangles show the plot of quantity $\langle \Delta^2 \rangle / (\sqrt{g} N^{1/4})^2$ against N . One can note that, $\langle \Delta^2 \rangle / (\sqrt{g} N^{1/4})^2$ tends to a constant value for large N and so, $\sqrt{\langle \Delta^2 \rangle}$ is of same order as quantity R . Other important thing to note is that classical space-time picture is maximally broken.

4.3 No SSB of Lorentz invariance

To check for the SSB of Lorentz invariance in the IIB model, we define a quantity J as

$$J = \frac{1}{D} I_{\mu\nu} I_{\mu\nu} - \left(\frac{1}{D} I_{\mu\mu} \right)^2, \quad (56)$$

where $I_{\mu\nu}$ is defined by

$$I_{\mu\nu} = \text{Tr}(A_\mu A_\nu) / N. \quad (57)$$

.

The quantity J represents the variation of the eigenvalues of $I_{\mu\nu}$.

If the quantity $\langle J \rangle / R^4$ is nonzero for large values of N , the Lorentz invariance is spontaneously broken. Therefore, $\langle J \rangle / (\sqrt{g} N^{1/4})^4$ is considered as an order parameter of the SSB of Lorentz invariance.

Fig. 8 shows that the order parameter goes to zero in the limit $N \rightarrow \infty$, which means that the Lorentz invariance is not spontaneously broken in the model.

4.4 Simulation Results

First, we set our parameters such that, the acceptance rate for the Leapfrog step is between 0.7 and 0.8 and, the action accumulates to some constant value. Then we simulate the model for $N = 4, 6, 8, 10$ by fixing $D (= 3, 4, 6, 10, 15, 20)$ and fit it with linear function.

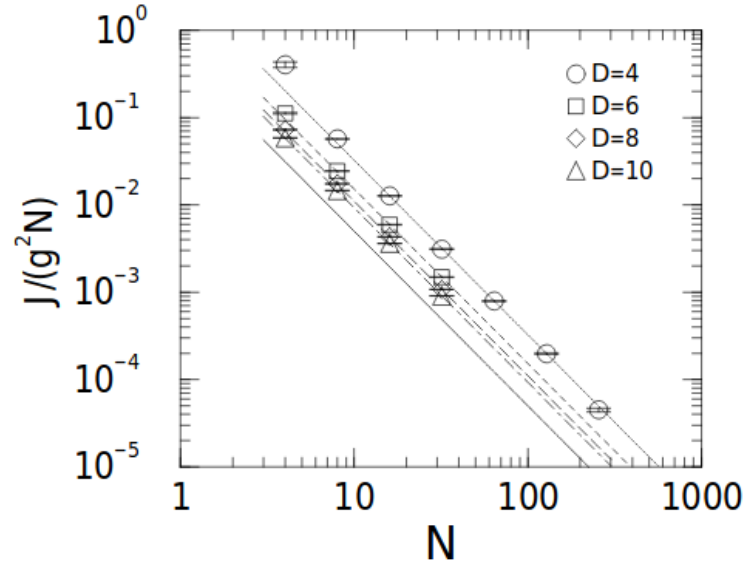


Figure 8: $\langle J \rangle / (\sqrt{g} N^{1/4})^4$ against N for $D = 4, 6, 8, 10$. The solid line represents the prediction for the $D \rightarrow \infty$ limit. This figure is taken from Ref. [1].

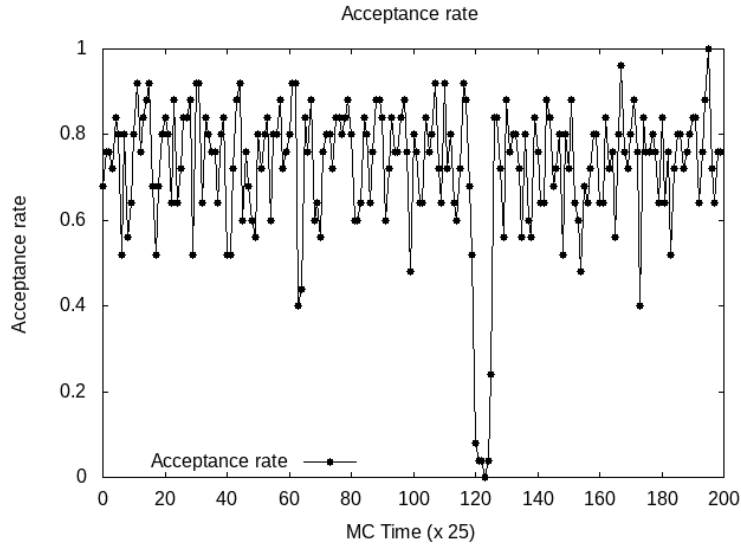


Figure 9: The acceptance rate against Monte Carlo time. We used $D = 4$ and $N = 4$ in the simulations.

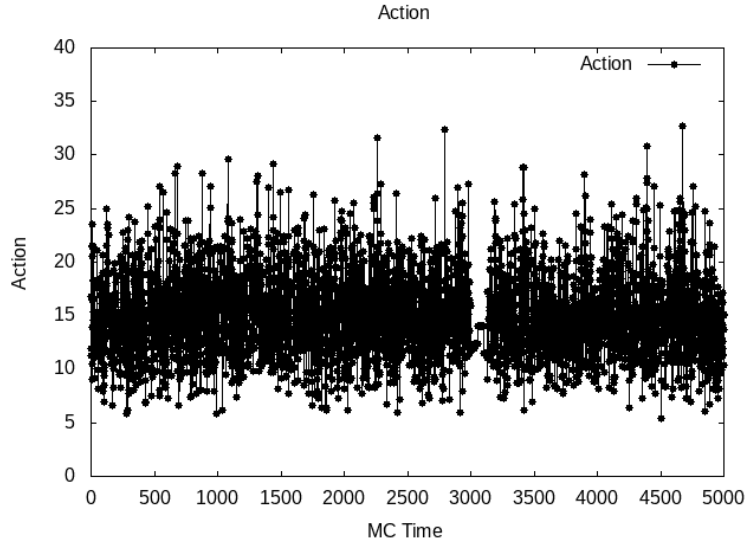


Figure 10: The value of the action against Monte Carlo time. We used $D = 4$ and $N = 4$ in the simulations.

Fig. 11 shows the logarithmic plot of $\langle J \rangle / (\sqrt{g} N^{1/4})^4$ with N

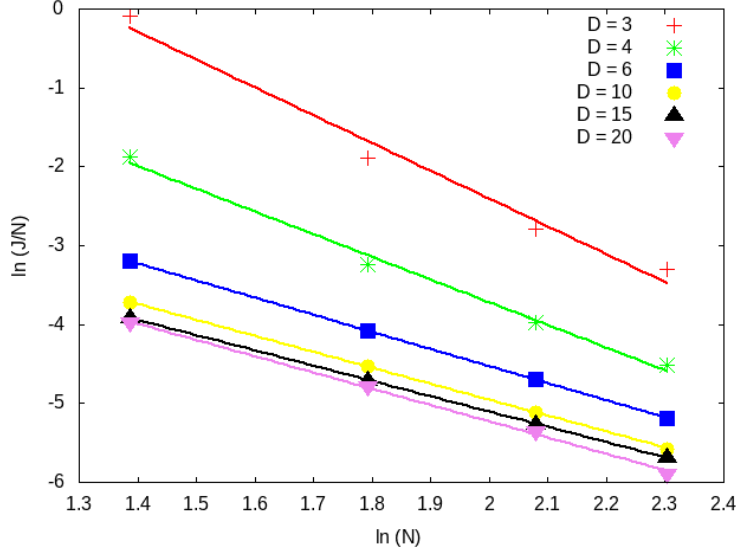


Figure 11: $\langle J \rangle / (\sqrt{g} N^{1/4})^4$ against N for $D = 3, 4, 6, 10, 15, 20$.

4.5 The Flowchart of the C++ code

Following Ref. [2] we show the flowchart of the C++ code we used in the simulations.

4.5.1 Description of Files Used

Below we provide the description of the C++ program files used in the simulations.

1. *read_param.cpp*: Reads in the simulation parameters from a data file called parameters.
2. *ssb_matrix.cpp*: The main file where all the functions are called and allowed to iterate.
3. *update.cpp*: Updates the field configurations based on HMC test.
4. *kinetic_energy.cpp*: Computes the kinetic energy term in the Hamiltonian.
5. *force.cpp*: Computes the bosonic contribution to the force.

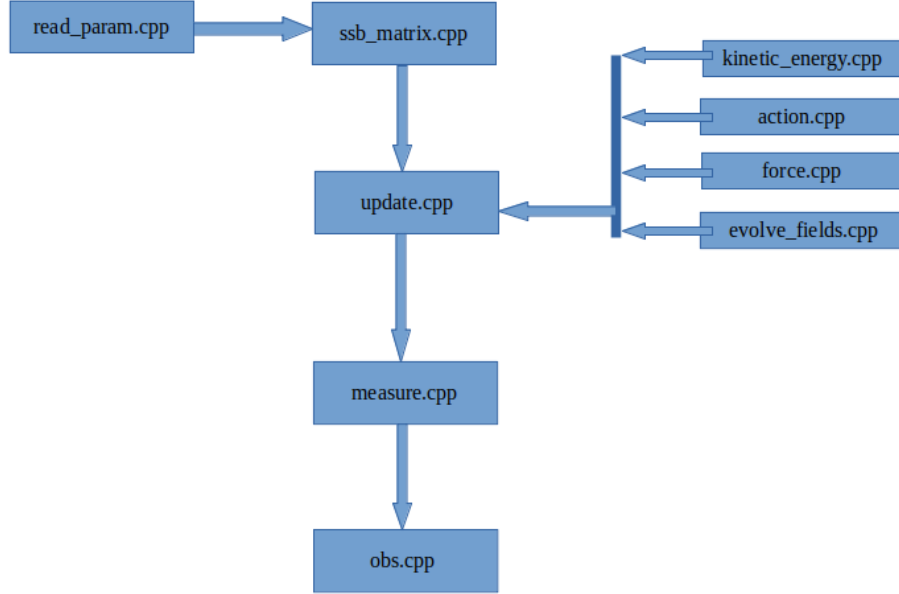


Figure 12: The flowchart of the C++ code used for the simulations of the matrix model.

6. *action.cpp*: Computes the total action.
7. *evolve_fields.cpp*: Leapfrog evolution algorithm.
8. *measure.cpp*: Performs the measurements on field configurations.
9. *obs.cpp*: Computes the observables.

5 Conclusions

We have successfully simulated a large- N matrix model using the HMC algorithm. We simulated the model for various values of D , the number of fields and N the rank of the matrix. We got the expected behavior for the quantity J for different values of N . We found that the quantity J decreases with an increase in the values of D and N , and goes to zero in $N \rightarrow \infty$. Thus, we showed that Lorentz invariance is not spontaneously broken.

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